

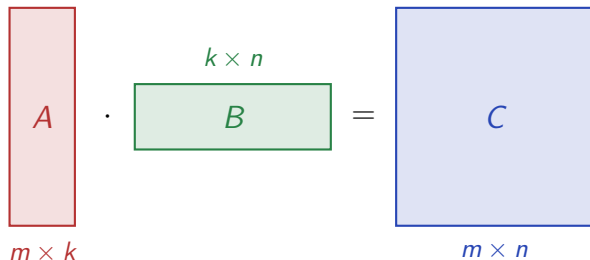
Cache-Optimal Tiling for Mixed-Precision Matrix Multiplication

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Matrix Multiply is Memory-Bound



for example:

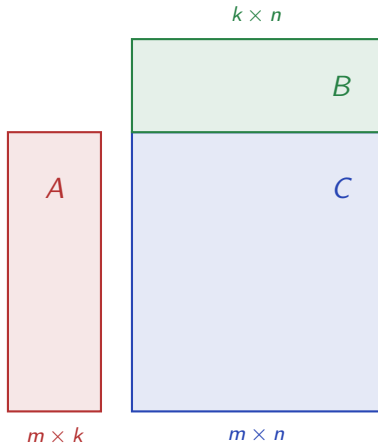
256×256 fp64: ≈ 0.5 MB

Typical L1 cache: 16–64 KB

$\Rightarrow 8 \times - 32 \times$ **larger than L1**

Fix: *tile* the loop — bring in one smaller C tile and keep it resident in L1.

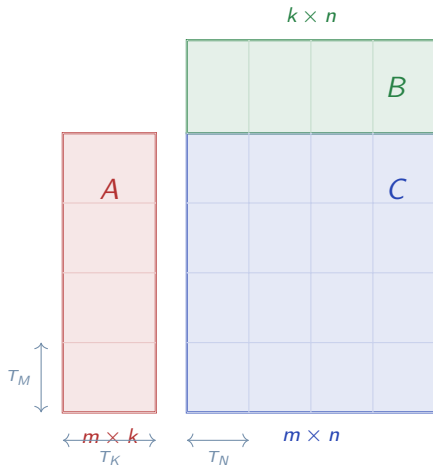
C-Stationary Tiling: Step by Step



$$C = A \cdot B$$

For large m, n, k — none fit in L1.

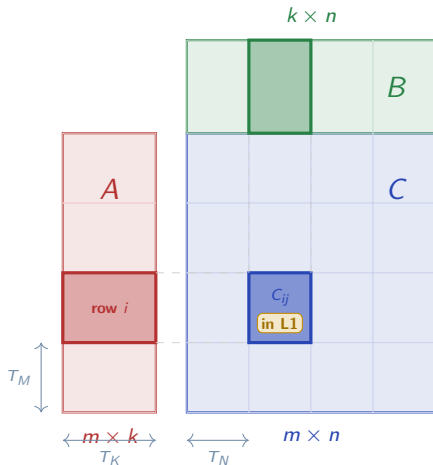
C-Stationary Tiling: Step by Step



Tile shape: $T_M \times T_N \times T_K$

Each matrix divided into a grid of tiles.

C-Stationary Tiling: Step by Step

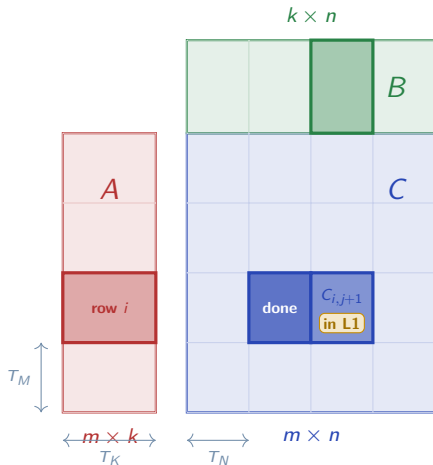


Fix C_{ij} in L1.

Stream A row-band and B col-band through it.

$$C_{ij} += A_{\text{row } i} \times B_{\text{col } j}$$

C-Stationary Tiling: Step by Step



C_{ij} done — move to B col $j+1$.

A row-band stays in L1.

Asymmetric Precision $\Rightarrow$ Asymmetric Tiles

$$\#A \text{ Bytes} = T_M \cdot k \cdot A_p$$

$$\#B \text{ Bytes} = T_N \cdot k \cdot B_p$$

$$\Rightarrow T = \frac{MN}{T_M T_N} (T_M \cdot k \cdot A_p + T_N \cdot k \cdot B_p)$$

$$T = MNK \left(\frac{A_p}{T_N} + \frac{B_p}{T_M} \right)$$

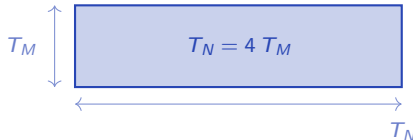
Find best ratio:

$$\Rightarrow \frac{T_N^*}{T_M^*} = \frac{A_p}{B_p} = \frac{1}{\rho}$$

$$\rho = 1 \text{ (fp32} \times \text{fp32)}$$

 T_N

$$\rho = \frac{1}{4} \text{ (fp32} \times \text{fp8)}$$



Same cache budget S — wider tile, better B reuse.

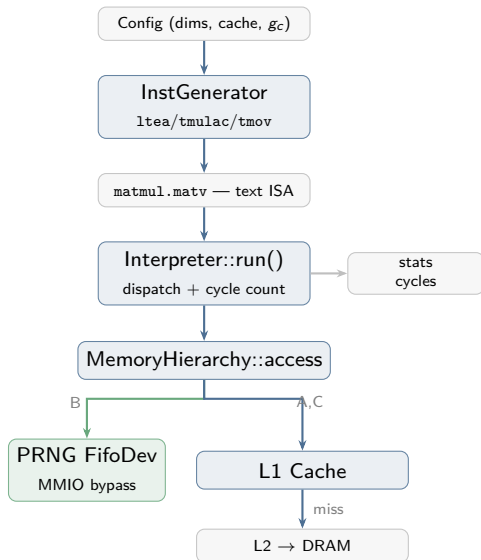
Default Hardware Configuration

L1 cache	16 KB, 64-byte lines, 4-cycle access
L2 cache	64 KB, 14-cycle access (off in the no-L2 runs)
DRAM	180-cycle access
Register size	$4 \times 4 \times 4$

The Simulator We Built

Three main components:

- **InstGenerator** — compiles tile ops to a text ISA (ltea/tmulac/tmov)
- **Interpreter** — dispatches instructions, counts cycles per access
- **MemoryHierarchy** — configurable L1/L2/DRAM with per-access tracing

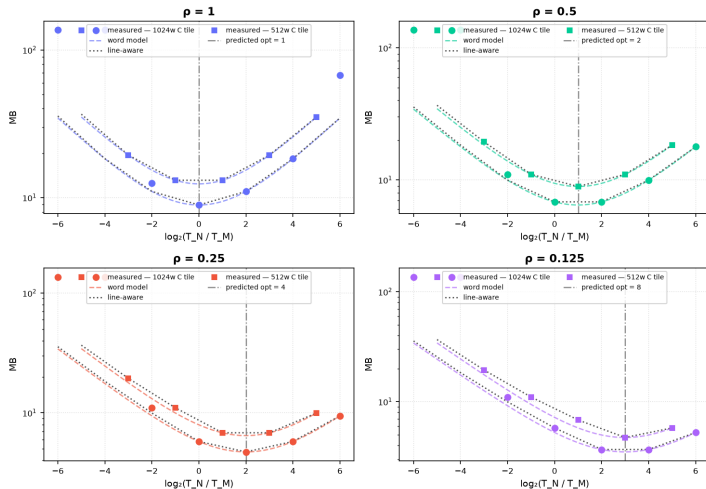


What the Model Assumes

- Tiles divide the matrix evenly, and register tiles divide the cache tile, so there are no partial edge tiles.
- A and C are stored in memory as row-major
- The C tile stays resident in L1 for the whole inner loop
- The way FIFO PRNG works is that: it uses seeds, which have smaller precision than elements of B and need to be taken from memory. We assume that this traffic is negligible and don't include it in our model.

Traffic Minimum at the Predicted Tile

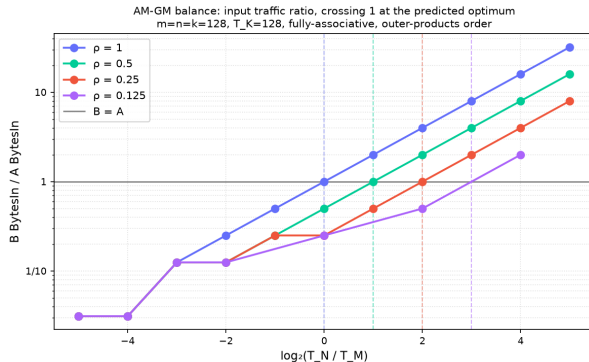
L1 reads (BytesIn) vs paper prediction — fully-associative cache
 $m=n=k=256$, $T_K=256$, outer-products order



Each curve shows L1 reads as a function of tile aspect ratio T_N/T_M .

Dashed line: paper's prediction $T_N/T_M = 1/\rho$.

B/A Balance: Exact Confirmation



Each curve is the ratio of B reads to A reads as T_N/T_M varies.

Traffic saved vs. square tile:

ρ	saving (empirical)	saving (theoretical)
1	0%	0%
0.5	0%	6%
0.25	18%	20%
0.125	36%	37%

Theoretical saving is given by the formula:

$$1 - \frac{2\sqrt{\rho}}{1 + \rho}$$

The paper minimizes *bytes transferred*:

$$T_{\text{traffic}} = MNK \left(\frac{A_P}{T_N} + \frac{B_P}{T_M} \right)$$

But cycles $\neq$ bytes.

We need a *cycle-accurate* model.

Enter the PRNG FIFO

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B elements are generated *on-chip* by a hardware random number generator, streamed directly into the compute unit.

B never touches the cache.

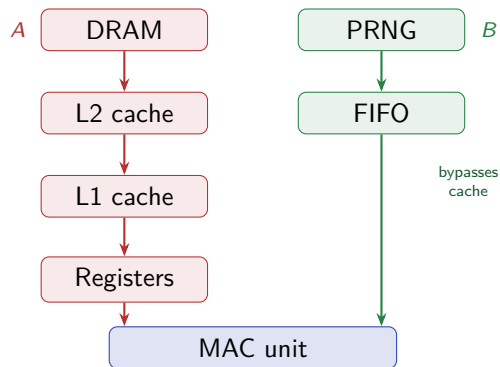
B has no memory address.

The B_P/T_M traffic term **vanishes**.

Instead, B introduces a new cost: on-chip *generation time*.

generation cost = g_c cycles / element

The PRNG FIFO: B Without Memory



Random matmul: $C = A \cdot B$ where B is a pseudo-random matrix generated on-chip.

Use cases: random projections, privacy-preserving ML, approximate algorithms.

Two independent bottlenecks

A-bound: load A from memory

B-bound: wait for PRNG (g_c cy/elem)

The PRNG FIFO: B Without Memory

Runtime = ?

The PRNG FIFO: B Without Memory

Runtime = whichever is slower

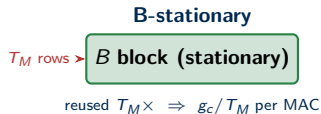
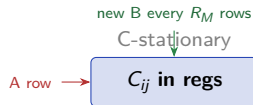
Which Loop Order Fits the FIFO?

Problem: B streams from the PRNG in a **fixed order** — you cannot skip or replay elements, so the loop must consume B as it is generated.

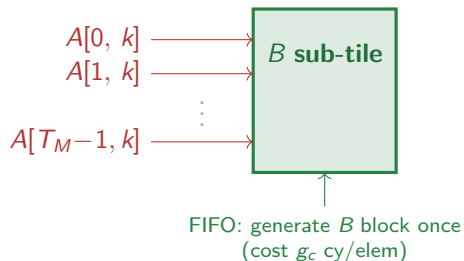
Two dataflows fit the stream. They differ in how many times each generated B element is reused:

Dataflow	In regs	B gen cost / MAC
C-stationary	C_{ij} tile	g_c / R_M
B-stationary	B_{ij} tile	g_c / T_M

$R_M =$ register-tile rows $\ll T_M$. C-stationary restarts the FIFO every R_M rows; B-stationary reuses each B element across the full tile height T_M .



B-Stationary: T_M -Fold Register Reuse



Loop order: B outer, A inner. Each B block is fetched *once*. $\rightarrow$ Each B element is used T_M times — generation cost amortized:

	$g_c=0$	$g_c=10$	$g_c=100$
B-stationary	ref	ref	ref
C-stationary	$1.7 \times \text{fstr}$	$1.4 \times \text{fstr}$	$7.1 \times \text{slwr}$

C-stationary wins while g_c is small (lower A-load cost), but B-stationary pulls ahead as generation dominates.

Naive Cycle Model: Two Cost Terms

Same reuse argument as traffic, but in **cycles**:

Naively assuming the two costs **add**:

$$\frac{T}{MNK} = \frac{C_A}{T_N} + \frac{g_c}{T_M}$$

FIFO is Async: Two Costs in Parallel

The FIFO generates B **in the background** while the core loads A from L1.

The two costs **overlap** — runtime is determined by whichever finishes last:

$$\frac{T}{MNK} = \max \left\{ \frac{C_A}{T_N}, \frac{g_c}{T_M} \right\}$$

Two regimes:

$g_c/T_M < C_A/T_N$: A-load bound

$g_c/T_M > C_A/T_N$: B-gen bound

Replacing C_A/T_N with Measured $\alpha(T_M, T_N)$

The C_A/T_N term assumes:

C_A = fixed cost per A element from L1
 T_N -fold reuse divides it out cleanly

Problem: actual cycles/MAC depends on tile shape through cache residency, line utilization, and access patterns — not a simple constant divided by T_N .

Solution:

$$\frac{T}{MNK} = \max \left\{ \alpha(T_M, T_N), \frac{g_c}{T_M} \right\}$$

Isolating α : Set $g_c = 0$

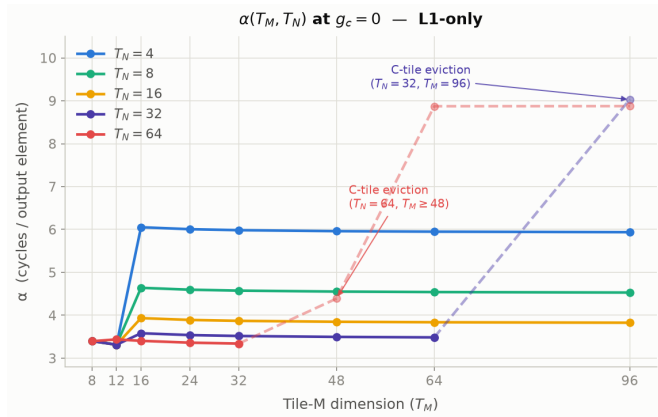
Key trick: set $g_c = 0$

The model collapses:

$$\frac{T}{MNK} = \max \left\{ \alpha(T_M, T_N), \frac{0}{T_M} \right\} = \alpha(T_M, T_N)$$

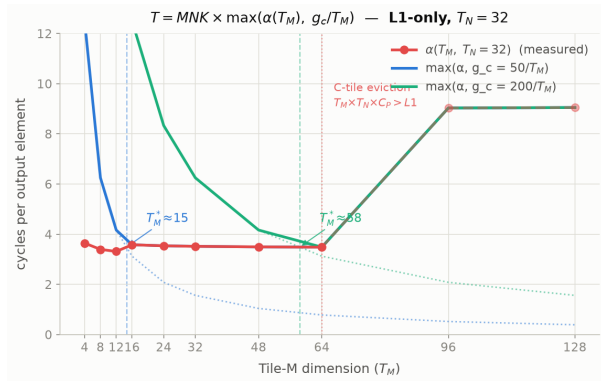
Calibration: Building the α Table

- Small T_N : α spikes when the A tile overflows L1
- Large T_N : α stays low until the C tile overflows L1
- α decreases with T_N (more column reuse)



Using the α Table: Predicting the Optimal Tile

The optimal tile is where the two costs are balanced: $\alpha(T_M^*, T_N^*) \approx g_c/T_M^*$.



Roofline Validation: Setup

Question:

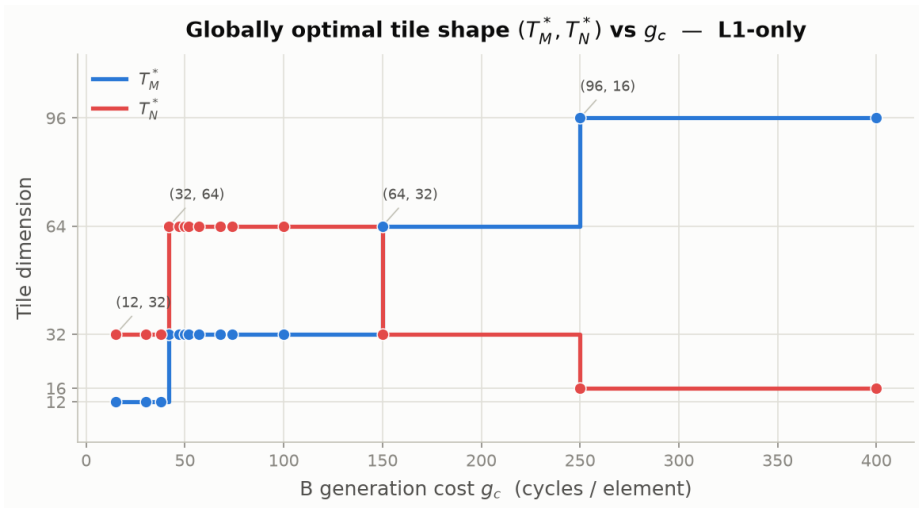
calibrate α once at $g_c = 0$; does the model predict the best tile correctly at any new g_c ?

Roofline Validation: Results

All 108 conditions	
(T_M^*, T_N^*) exact	108 / 108 (100%)

100% exact match across all hardware configurations and g_c values.

Impact of Tile Selection: Optimal vs. Square



Impact of Tile Selection: Optimal vs. Square

Question:

Does choosing the right (T_M, T_N) actually matter?

Comparing:

- Optimal asymmetric tile (T_M^*, T_N^*)
- Square tile $(T_M = T_N, \text{fixed})$

Speedup from correct tile:

- $g_c = 100$: **20–65%** faster
- $g_c = 250$: **~85%** faster
- $g_c = 400$: **up to 90%** faster

Impact of Tile Selection: Optimal vs. Square

L1-only — empirically measured performance (E8 sweep)

